

The q -integer $[3]_q$ as the engine of Theorem B, and the polynomial-vs.-multiset-positivity gap

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Abstract

The May 6 day-session produced five writeups establishing Theorems A, B, and C at the multiset level, the path-level Theorem B with explicit $n = 5$ injection, the refutation of Open Question Q3, and the structural reduction of the sign-positivity lemma to two single-component multiplicity bounds. This evening note adds one cleanly proved observation that ties them together. The sole “perturbation” separating atom_{RTL} from $A_{(3,2)}$ is the q -integer multiple $2q[3]_q$, where $[3]_q := 1 + q + q^2$. Convolving this into the $n = 6$ identity gives

$$A_{(3,2,1)}(q) = A_{(3,1,1)}(q) A_{(3,2)}(q) + 2q[3]_q A_{(3,1,1)}(q),$$

a polynomial decomposition whose perturbation $2q[3]_q A_{(3,1,1)}$ is itself palindromic, non-negative, and factors cleanly. The same perturbation, however, has *signed* canonical σ_1 -G1 multiplicity $(0, +2, +2, -4)$ on B_6 . The polynomial is non-negative but not multiset-positive in the canonical basis. This is exactly the obstruction to a direct multiset interpretation of the perturbation, and is the precise structural origin of the “hidden” atom atom_{RTL} . We take the result as a clean structural cap on the day’s program: the gap between polynomial- and multiset-positivity of the perturbation $2q[3]_q A_{(3,1,1)}$ is what forces the rewrite into atom_{RTL} , and is what makes Theorem B a genuine structural statement rather than a polynomial identity.

1 Setup

We adopt the framework of `2026-05-06-twin-multiset.tex`, `2026-05-06-theorem-B-multiset.tex`, and `2026-05-06-sign-positivity-structural.tex`. Recall the canonical σ_1 -G1 basis $B_{2k} = \{q^c(1+q)^{2k-2c} : c = 0, \dots, k\}$, which is $\mathbb{Q}[q]$ -linearly independent. Each cell-character $A_\lambda(q)$ admits a unique non-negative integer expansion in B_{2m_λ} , with multiplicity vector M_λ , and multiplication of polynomials becomes convolution of multiplicity vectors. The atoms relevant here:

$$M_{(3,1,1)} = (1, 2), \quad M_{(3,2)} = (1, 5, 3), \quad M_{(4,1)} = (1, 3, 5), \quad M_{(3,2,1)} = (1, 9, 15, 2), \quad M_{(5,1)} = (1, 3, 9, 14). \quad (1)$$

The polynomial $[k]_q := 1 + q + \dots + q^{k-1}$ is the q -integer of order k . We will use $[3]_q = 1 + q + q^2$ throughout.

2 The q -integer identity

Lemma 1 ($2q$ -difference, restated).

$$A_{(3,2)}(q) - A_{(4,1)}(q) = 2q \cdot [3]_q.$$

Proof. By direct expansion of $M_{(3,2)} - M_{(4,1)} = (0, +2, -2)$ on B_4 :

$$2q(1+q)^2 - 2q^2 = 2q[(1+q)^2 - q] = 2q[1+q+q^2] = 2q[3]_q.$$

The middle equality is $(1+q)^2 = 1+2q+q^2 = [3]_q + q$. □

Lemma 2 (Multiplicity vector of $[3]_q \cdot A_{(3,1,1)}$)

$$[3]_q A_{(3,1,1)}(q) = 1 + 5q + 6q^2 + 5q^3 + q^4.$$

The polynomial is palindromic of degree 4, with all coefficients non-negative.

Proof. $A_{(3,1,1)}(q) = (1+q)^2 + 2q = 1 + 4q + q^2$. Then

$$(1+q+q^2)(1+4q+q^2) = 1 + (1+4)q + (1+4+1)q^2 + (4+1)q^3 + q^4 = 1 + 5q + 6q^2 + 5q^3 + q^4.$$

Palindromicity is by inspection. □

Theorem 3 (Polynomial decomposition of $A_{(3,2,1)}$).

$$A_{(3,2,1)}(q) = A_{(3,1,1)}(q) A_{(3,2)}(q) + 2q [3]_q A_{(3,1,1)}(q),$$

and equivalently

$$A_{(5,1)}(q) = A_{(3,1,1)}(q) A_{(4,1)}(q) - 2q [3]_q A_{(3,1,1)}(q).$$

The perturbation polynomial $2q [3]_q A_{(3,1,1)}(q) = 2(q + 5q^2 + 6q^3 + 5q^4 + q^5)$ is palindromic, non-negative, and supported on $\bar{c} = 1, \dots, 5$ (degree 1 through 5 in q).

Proof. The multiset Theorem B (Theorem 3.1 of 2026-05-06-theorem-B-multiset.tex) is $M_{(3,2,1)} = M_{(3,1,1)} * \text{atom}_{\text{RTL}}$ where $\text{atom}_{\text{RTL}} = 2M_{(3,2)} - M_{(4,1)}$. Translating to polynomials,

$$A_{(3,2,1)} = A_{(3,1,1)} \cdot (2A_{(3,2)} - A_{(4,1)}) = A_{(3,1,1)} A_{(3,2)} + A_{(3,1,1)} (A_{(3,2)} - A_{(4,1)}).$$

By Lemma 1, $A_{(3,2)} - A_{(4,1)} = 2q[3]_q$, so the second summand is $A_{(3,1,1)} \cdot 2q[3]_q$. Multiplying out using Lemma 2: $2q[3]_q A_{(3,1,1)}(q) = 2q(1 + 5q + 6q^2 + 5q^3 + q^4) = 2q + 10q^2 + 12q^3 + 10q^4 + 2q^5$, palindromic.

The $A_{(5,1)}$ -form is analogous, using $A_{(5,1)} = A_{(3,1,1)}(2A_{(4,1)} - A_{(3,2)})$ and the negative of the same identity. □

Remark 4 (Numerical check). Direct expansion:

$$\begin{aligned} A_{(3,1,1)}(q) A_{(3,2)}(q) &= (1 + 4q + q^2)(1 + 9q + 19q^2 + 9q^3 + q^4) \\ &= 1 + 13q + 56q^2 + 94q^3 + 56q^4 + 13q^5 + q^6, \\ 2q[3]_q A_{(3,1,1)}(q) &= 2q + 10q^2 + 12q^3 + 10q^4 + 2q^5, \\ \text{sum} &= 1 + 15q + 66q^2 + 106q^3 + 66q^4 + 15q^5 + q^6 = A_{(3,2,1)}(q). \checkmark \end{aligned}$$

3 The polynomial-vs.-multiset positivity gap

The polynomial $2q[3]_q A_{(3,1,1)}(q)$ has all non-negative coefficients (it is a positive polynomial). But its expansion in the canonical σ_1 -G1 basis B_6 involves a negative entry. This is the structural origin of why atom_{RTL} exists as a hidden multiset rather than as a direct “ $A_{(3,1,1)} p(q)$ ” multiset for some positive multiset p .

Lemma 5 (Signed B_6 -decomposition of the perturbation). *On $B_6 = \{(1+q)^6, q(1+q)^4, q^2(1+q)^2, q^3\}$, the perturbation has unique signed expansion*

$$2q[3]_q A_{(3,1,1)}(q) = 0 \cdot (1+q)^6 + 2 \cdot q(1+q)^4 + 2 \cdot q^2(1+q)^2 + (-4) \cdot q^3.$$

That is, signed B_6 -multiplicity $(0, +2, +2, -4)$.

Proof. By Theorem 3, $A_{(3,2,1)} - A_{(3,1,1)}A_{(3,2)} = 2q[3]_q A_{(3,1,1)}$, which corresponds on B_6 to $M_{(3,2,1)} - M_{(3,1,1)} * M_{(3,2)} = (1, 9, 15, 2) - (1, 2) * (1, 5, 3) = (1, 9, 15, 2) - (1, 7, 13, 6) = (0, +2, +2, -4)$.

Verifying the polynomial expansion: $0 + 2q(1+q)^4 + 2q^2(1+q)^2 - 4q^3$.

$$\begin{aligned} 2q(1+q)^4 &= 2(q + 4q^2 + 6q^3 + 4q^4 + q^5) = 2q + 8q^2 + 12q^3 + 8q^4 + 2q^5, \\ 2q^2(1+q)^2 &= 2(q^2 + 2q^3 + q^4) = 2q^2 + 4q^3 + 2q^4, \\ -4q^3 &= -4q^3. \end{aligned}$$

Sum: $2q + (8+2)q^2 + (12+4-4)q^3 + (8+2)q^4 + 2q^5 = 2q + 10q^2 + 12q^3 + 10q^4 + 2q^5$, matching Theorem 3. The expansion is unique by the $\mathbb{Q}[q]$ -linear independence of B_6 . $\square$

Corollary 6 (The polynomial is positive but not multiset-positive). *The polynomial $2q[3]_q A_{(3,1,1)}(q)$ has all non-negative coefficients in $\mathbb{Z}[q]$, hence is positive in the polynomial sense. But its canonical σ_1 -G1 multiplicity vector $(0, +2, +2, -4)$ on B_6 has a negative entry at $\bar{c} = 3$. So the perturbation is not a non-negative element of the σ_1 -G1 multiset cone over B_6 .*

Proof. Both statements are explicit: the polynomial coefficients $(0, 2, 10, 12, 10, 2, 0)$ are non-negative in $\mathbb{Z}[q]$; the B_6 -multiplicity $(0, +2, +2, -4)$ has a negative third entry. $\square$

Remark 7 (Why this matters). This corollary is the structural reason why Theorem B at the multiset level is non-trivial. If the perturbation $2q[3]_q A_{(3,1,1)}$ had non-negative B_6 -multiplicity, then $M_{(3,2,1)}$ would simply be $M_{(3,1,1)} * M_{(3,2)}$ plus a non-negative multiset, with no need for the hidden atom rewrite $\text{atom}_{\text{RTL}} = 2M_{(3,2)} - M_{(4,1)}$. But the perturbation has a -4 at $\bar{c} = 3$, so this naive “add-positive” reading fails. The rewrite $M_{(3,2,1)} = M_{(3,1,1)} * \text{atom}_{\text{RTL}}$ absorbs the perturbation by introducing $-M_{(4,1)}$ inside atom_{RTL} .

Concretely: at $\bar{c} = 3$, $M_{(3,1,1)} * M_{(3,2)} = 6$, but $M_{(3,2,1)} = 2$. We need to *remove* 4 paths-worth at this grade. In multiset language this is impossible (you cannot have a negative number of paths). The trick is to rewrite using the doubled atom: $M_{(3,1,1)} * 2M_{(3,2)} = 12$, and $M_{(3,1,1)} * M_{(4,1)} = 10$, with difference $12 - 10 = 2$. This is the multiset Theorem B (or, the path-level Theorem B as a containment-and-complement).

So the gap between polynomial- and multiset-positivity, exhibited quantitatively by Lemma 5, is the precise structural origin of atom_{RTL} as a hidden multiset.

4 What the $[3]_q$ factor suggests

The factorization in Theorem 3 isolates the q -integer $[3]_q$ as the sole non-trivial factor in the perturbation. We make a few observations.

$[3]_q$ vs $A_{(2,1)}$ at S_3

The polynomial $[3]_q = 1 + q + q^2$ is the q -character of the trivial representation of S_2 induced to S_3 , equivalently the q -Hilbert series of projective space $\mathbb{P}^2$ (or of any flag variety with three Bruhat cells of consecutive lengths). The post-prefix σ_1 -G1 atom $A_{(2,1)}(q)$ at S_3 has the same degree (degree 2) and is also palindromic. We have not verified whether $A_{(2,1)}(q) = [3]_q$ at S_3 in this note (the comparison requires recomputing $A_{(2,1)}$ in the σ_1 -G1 post-prefix convention), but the coincidence is suggestive: if it holds, then $2q[3]_q = 2q A_{(2,1)}(q)$ and Theorem 3 reads

$$A_{(3,2,1)}(q) = A_{(3,1,1)}(q) A_{(3,2)}(q) + 2q A_{(3,1,1)}(q) A_{(2,1)}(q).$$

This would give the perturbation a clean partition-theoretic identity.

Open question: which other partition pairs admit a q -integer perturbation?

The pair $((3, 2), (4, 1))$ at S_5 is special because $A_{(3,2)} - A_{(4,1)} = 2q[3]_q$ is a clean q -integer multiple, with $[3]_q$ palindromic of degree 2 ($n = 3$ in the q -integer family). A natural question is: which other partition pairs (λ, μ) at S_n have $A_\lambda - A_\mu = c \cdot q^a \cdot [k]_q$ for some positive integer c , non-negative integer a , and integer $k \geq 2$? Each such pair would give an analogous Theorem-B-like multiset rewrite. The next session should test this systematically at $n = 6, 7$ pairs.

5 What this does and does not prove

What this proves. The clean polynomial decomposition $A_{(3,2,1)} = A_{(3,1,1)}A_{(3,2)} + 2q[3]_qA_{(3,1,1)}$, the corresponding $A_{(5,1)} = A_{(3,1,1)}A_{(4,1)} - 2q[3]_qA_{(3,1,1)}$, and the explicit signed B_6 -multiplicity vector $(0, +2, +2, -4)$ of the perturbation. The polynomial is non-negative but not multiset-non-negative.

What this does not prove. It does *not* construct a structural W -graph-natural injection of paths, does not prove canonicity, and does not prove either of the two open empirical identities $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$ (Theorem 4 of 2026-05-06-paths-explicit-and-Q3-refuted.tex) or any generalization beyond the specific $n = 5, 6$ pair.

Status of the May 6 program at end of evening. The polynomial identities of Theorems A, B, C are all verified. Their multiset upgrades are all proved (Theorem B with hidden $\text{atom}_{\text{RTL}} = (1, 7, 1)$ and $\text{atom}_{\text{LTR}} = (1, 1, 7)$ as the structural inner factors). The path-level Theorem B is proved as an existence-of-injection statement (with an explicitly tabulated lex injection). Open Question Q3 of 2026-05-06-theorem-B-multiset.tex is refuted; in its place we have the empirical identity $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$ from 2026-05-06-paths-explicit-and-Q3-refuted.tex, whose structural meaning is open. The sign-positivity lemma is proved and its W -graph-natural status is reduced to two single-component multiplicity bounds (Theorem 1 of 2026-05-06-sign-positivity-structural.tex). The current note adds the $[3]_q$ -isolation of the perturbation and the polynomial-vs.-multiset positivity gap as the precise structural fact behind the day's program.

Computational verification

- $A_{(3,2)}(q) - A_{(4,1)}(q) = 2q(q^2 + q + 1) = 2q[3]_q$ (April 29 polynomial residual zero; Lemma 3 of 2026-05-06-theorem-B-multiset.tex; Lemma 1 above).
- $[3]_q A_{(3,1,1)}(q) = 1 + 5q + 6q^2 + 5q^3 + q^4$ (direct expansion; Lemma 2).
- $A_{(3,1,1)}(q) A_{(3,2)}(q) = 1 + 13q + 56q^2 + 94q^3 + 56q^4 + 13q^5 + q^6$ (Remark 4).

- $A_{(3,2,1)}(q) = 1 + 15q + 66q^2 + 106q^3 + 66q^4 + 15q^5 + q^6$ (2026-05-06-twin-multiset.tex).
- Difference $A_{(3,2,1)} - A_{(3,1,1)}A_{(3,2)} = 2q + 10q^2 + 12q^3 + 10q^4 + 2q^5 = 2q[3]_q A_{(3,1,1)}$ (Theorem 3).
- B_6 -decomposition of the perturbation: $(0, +2, +2, -4)$ (Lemma 5).